

# Assessment-1

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explain how human perceptual processing models and gestalt principle influence the effectiveness of data visualization  
discuss with suitable example how visualization designer can minimize information overload and maximize information clarity using concept such as Gibson affordance theory, data abstraction, and appropriate dataset representation

## Introduction

Data visualization is one of the most powerful tools in modern communication because it transforms raw data into meaningful interpretable patterns. Its effectiveness depends on how well it aligns with human perception and cognition.

By applying perceptual principles like gestalt principles and concepts like Gibson's affordance theory, data and designer proper dataset representation, designer can create clear, intuitive and user friendly visualization that reduce confusion and enhance understanding.

Human perceptual processing model in visualization  
Human perceptual models explain how people quickly intercept visual data by recognizing patterns, colors, shapes and contrasts faster than text.

### Pre-attentive processing

The brain instantly notices features like color size or orientation. Example a red bar in chart of blue bar draws immediately attention.

working memory limits.

since human can only process few chunks of information at once ( $7 \pm 2$ ) visualization should summarize data.



result principles and data visualization

result psychology explain how human naturally perceive and grouping. These principles are essential in visualization design because charts or dashboard proximity:

elements that are close together are perceived as belonging to the same group.

similarity:

object with similar shapes color or sizes are seen as part of the same category

continuity:

The human eye prefers continuous lines and curves line chart are effective because viewer naturally follow trend along a smooth path.

closure

humans tend to fill in gaps to perceive a complete shape

Minimizing

information overload and maximizing clarity

1. Gibson's

Affordance Theory suggest that objects have inherent visualization. This translate to intuitive

inference

design where the user instantly understand how to interact with or interpret the chart.

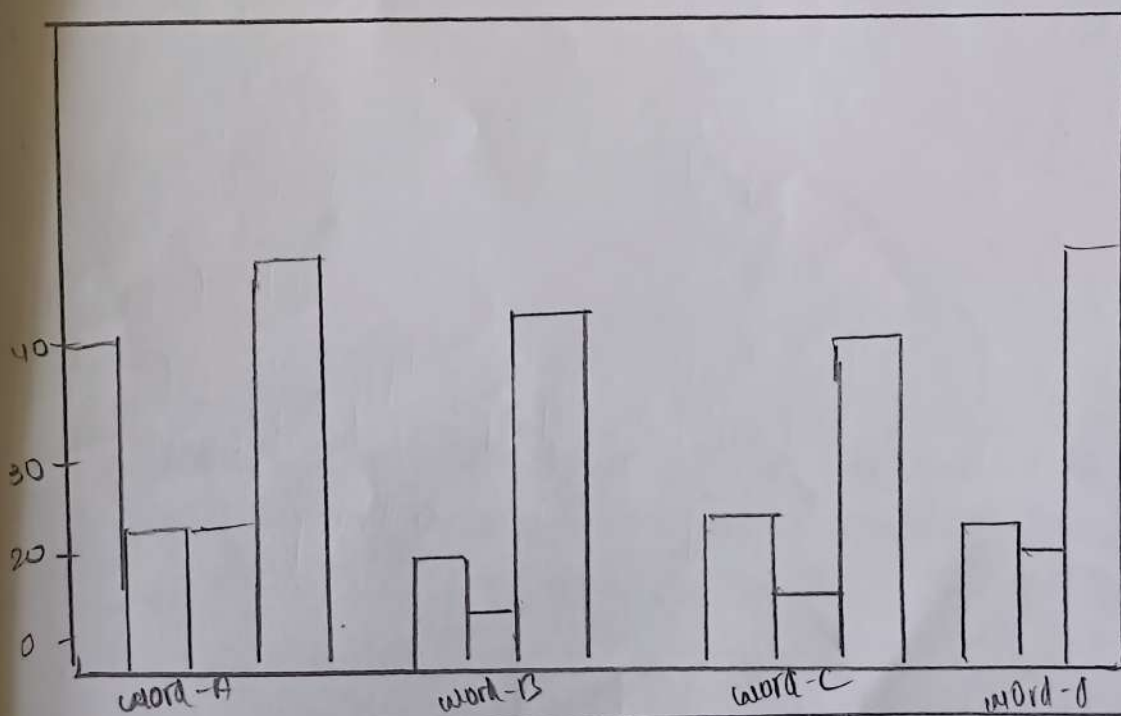
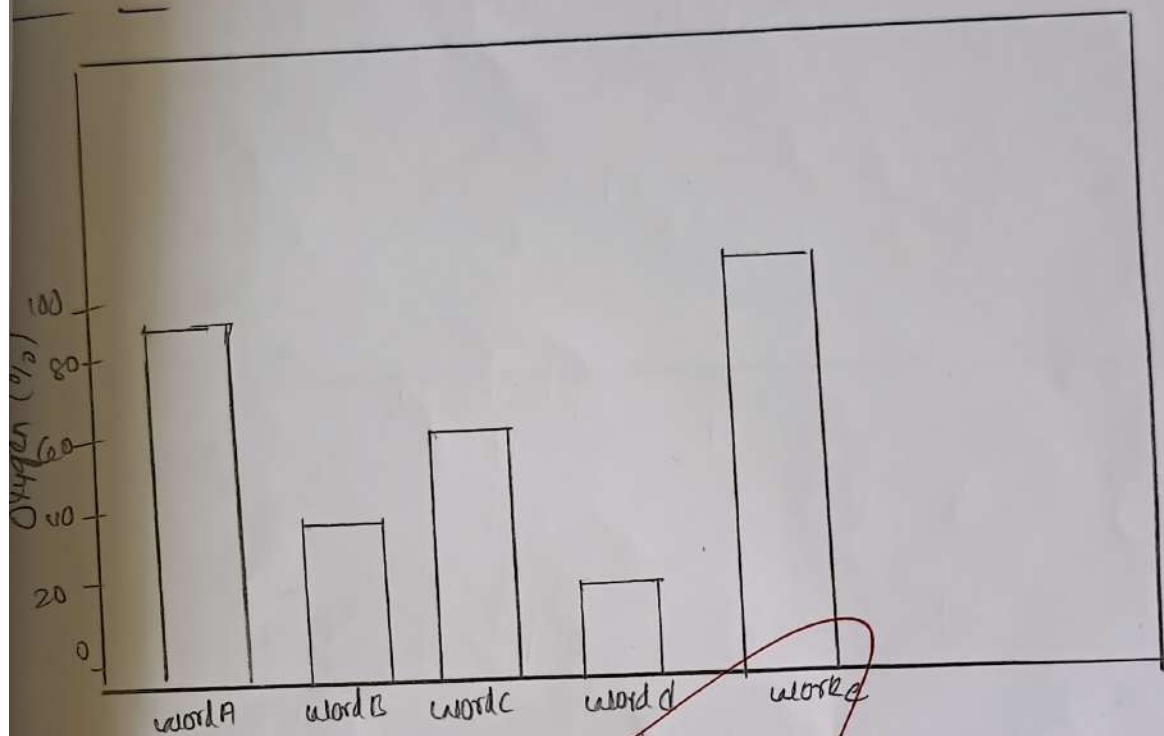
2. Data Abstraction

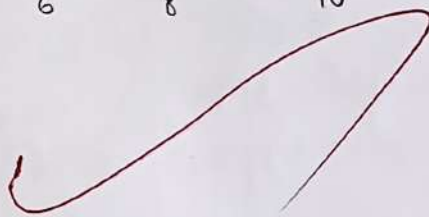
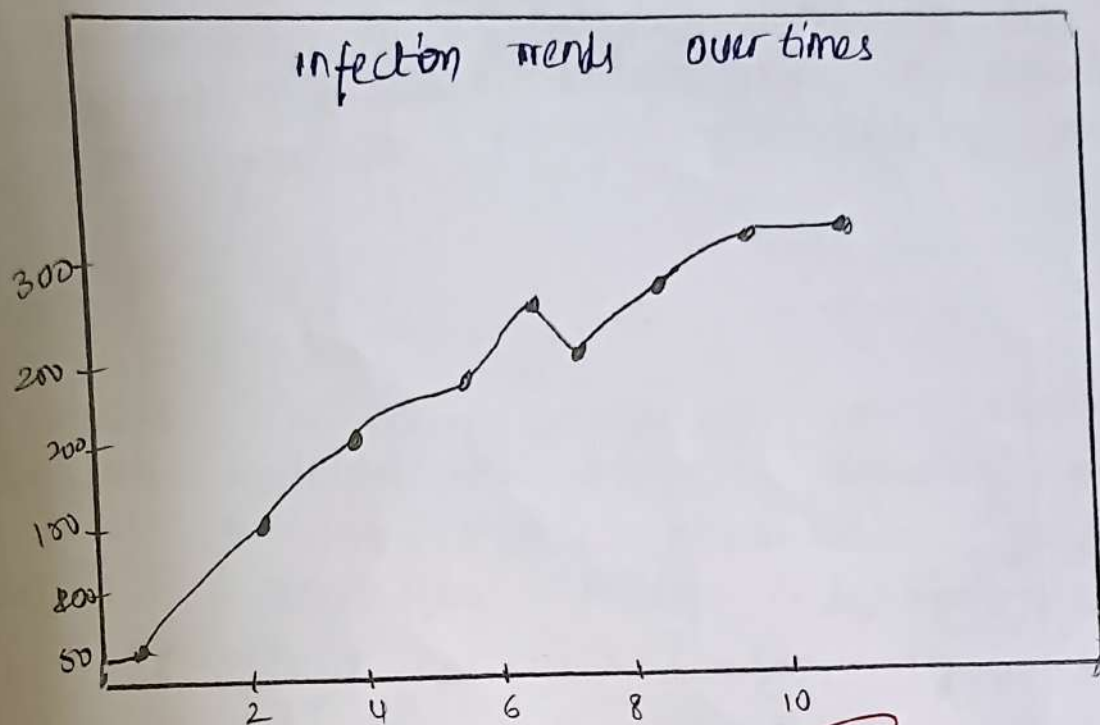
Raw data is often too large and complex for direct visualization. Data abstraction helps by reducing complexity while preserving meaning.

## practical example

1) a hospital dashboard during a pandemic, using  
color to highlight low oxygen level grouping  
data by word adding interactive filter and chose  
oper visuals helps doctor quickly intercept data.

### Bar chart





~~Conclusion:~~

Therefore the effectiveness of visualization depend on how human perceive information using perceptual model, gestalt principle, affordance theory



With the help of suitable dataset Compare and contrast different visualization techniques used in univariate Bivariate and multivariate analysis. Explain how the choice of visualization depends on the type of provide at least one practical example for example analysis type

## Introduction

Data analysis is often categorized into univariate Bivariate and multivariate analysis depending on the number of variable studied. Visualization plays a vital role in identifying pattern, relationships and anomalies. The choice of visualization depends on:

1. Nature of data: Categorical vs Continuous
2. Number of variable: one, two or more than two.
3. Objective of analysis: distribution, comparison or relationship.

## Univariate Analysis

Univariate analysis involves analyzing a single variable at a time to understand its distribution, central tendency, and spread.

Suitable visualization techniques

Bar chart: used when the variable is Categorical

Pie chart: used when the variable is partitioned into the form of sectors.

Examples:

1) Indian Census (male and female Categorized by bar chart)

2) Monthly Expenditure (pie chart) (various food saving etc);

## Bivariate Analyze

Bivariate Analyze deals with relationship between two variable. The aim is to identify correlation, trend, or different between them.

Visualization techniques:

Scatterplot with Fit Line: used for two continuous variables.

Side by side Boxplot:

used when one variable is categorical and the other is continuous.

Bubble chart:

similar to a scatterplot but with a third variable represented by bubble size or color.

Example:

Consider a dataset of patients in a hospital with variable age (continuous), blood pressure and cholesterol. A heat map can show correlation between age, blood pressure and cholesterol. A bubble chart could plot age vs cholesterol while using bubble size to represent blood pressure.

## Conclusion

Therefore univariate, bivariate and multivariate visualization serve different purpose distribution, relationships and complex interaction.

Choosing the right chart type based on data type and help turn raw data into insights supporting better decisions.